

1. Administrative & Study Identification

Full Study Title: A Study to Investigate the Efficacy and Safety of Crizanlizumab (5 mg/kg) Compared With Placebo in Adolescent and Adult Sickle Cell Disease Patients Who Experience Frequent Vaso-Occlusive Crises (SPARKLE) **Short Title/Acronym:** SPARKLE **Protocol Number:** CSEG101A2303 **NCT Number:** NCT06439082 **Trial Status:** Recruiting **Sponsor/Company Name:** Novartis Pharmaceuticals **Investigational Product:** Crizanlizumab (5 mg/kg) **Phase of Development:** Phase III **Disease Area:** Sickle Cell Disease **Medical Monitor:** Novartis Clinical Operations Lead [Contact Information on File]

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the efficacy and safety of crizanlizumab 5 mg/kg versus placebo, with or without hydroxyurea/hydroxycarbamide (HU/HC) therapy, in adolescent and adult Sickle Cell Disease (SCD) patients with frequent vaso-occlusive crises. **Secondary Objectives:** To evaluate the annualized rate of all VOCs (both HCP-managed and self-managed), annualized rate of VOC by subtype of management, time to first HCP-managed VOC, proportion of participants free from HCP-managed VOCs, duration of HCP-managed VOCs, and safety assessments including adverse events of special interest (infusion-related reactions, infections).

2.2 Background & Rationale To address the conflicting findings from previous crizanlizumab trials (SUSTAIN and STAND) where healthcare access disruptions during the COVID-19 pandemic may have led to underreporting of VOCs. The SPARKLE study implements a redefined VOC endpoint (pain crisis lasting ≥ 4 hours, treated per local standard-of-care) and utilizes a novel cloud-based application to capture real-time, self-managed VOCs to confirm the clinical efficacy of crizanlizumab.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** 2:1 ratio to crizanlizumab 5 mg/kg or placebo. Central randomization is stratified by concomitant HU/HC usage (yes/no) and region (South America, North America, and sub-Saharan Africa). **Status:** Recruiting

3.2 Planned Enrollment & Duration Target **\$N\$:** 315 participants
Number of Sites: Multicenter (Global, including South America, North America, and sub-Saharan Africa) **Estimated Study Start:** 24-Oct-2024 **Estimated Primary Completion:** 19-Apr-2030

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adolescent (12 to <18 years old) and adult (≥ 18 years) patients. 2. Confirmed diagnosis of SCD by Hb electrophoresis or high-performance liquid chromatography (HPLC). All SCD genotypes are eligible. 3. Experienced 4 to 12 VOCs that are HCP-managed (including management at a health care facility or via remote consultation) within the 12 months prior to the screening visit. 4. If the participant is on HU/HC, they must be taking it for at least 6 months and at a stable dose for at least 3 months prior to the Screening visit. If receiving an erythropoietin stimulating agent, the participant must have been receiving the drug for at least 6 months prior to the screening visit. Participants who have not been receiving HU/HC and/or erythropoietin stimulating agents must not have received them for at least 6 months prior to the screening visit.

4.2 Exclusion Criteria 1. Fewer than 4 or more than 12 VOCs that are HCP-managed within the 12 months prior to screening. 2. History of stem cell transplant and/or gene therapy. 3. Received blood products within 30 days prior to Week 1 Day 1 dosing. 4. Documented history of a clinical stroke, intracranial hemorrhage, or uninvestigated neurologic finding within the past 12 months before screening (silent infarct only present on imaging is not excluded). 5. Participating in a chronic transfusion program (pre-planned series of transfusions for prophylactic purposes) and/or planning to undergo an exchange transfusion during the duration of the study; episodic transfusion in response to worsened anemia or VOC is permitted. 6. Contraindication or hypersensitivity to any drug or metabolites from similar class as study drug or to any excipients of the study drug formulation. History of severe hypersensitivity reaction to other monoclonal antibodies.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Crizanlizumab 5 mg/kg or matching Placebo. **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** 52-week treatment period.

5.2 Concomitant & Prohibited Therapy Permitted: Standard of care therapy, including stable doses of hydroxyurea/hydroxycarbamide (HU/HC) therapy. Episodic transfusions in response to worsened anemia or VOCs. **Prohibited:** Initiation of HU/HC and/or erythropoietin stimulating agents within 6 months prior to

screening. Chronic transfusion programs or planned exchange transfusions are excluded.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Hb electrophoresis/HPLC, confirmation of baseline VOC rate
Baseline	Day 1	Randomization, First IV Infusion
Interim	Weeks 1 to 52	IV infusions, Safety Labs, Efficacy Assessment, remote tracking of VOCs via cloud-based app
End of Study	Follow-up	Final vital signs, Exit interview, IP Accountability (+27 days post last dose)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Annualized rate of VOC events that are HCP-managed (including VOCs leading to management at a health care facility or remote consultation) over the 52-week treatment period. A VOC is defined as an acute onset of pain for which there is no other medically determined explanation other than vaso-occlusion, lasting for at least 4 hours, which is treated as per local guidelines. Acute chest syndrome (ACS), priapism, and hepatic or splenic sequestration will be considered VOC in this study. **Measurement Method:** Clinical reporting calculation: $(\text{Number of VOC events} \times 365) / (\text{number of days in the observation period})$. Observation period = time from date of randomization to minimum of (last dose date until treatment discontinuation + 27 days, date of initiation or discontinuation of HU/HC or L-Glutamine [or other therapies to treat SCD and/or to prevent/reduce VOCs], date of randomization + 365 days).

7.2 Secondary & Exploratory Endpoints * Annualized rate of all VOCs including VOCs that are HCP-managed (either at a health care facility or via remote consultation) as well as those that are self-managed without recommendations from HCP captured via a cloud-based app. * Annualized rate of VOC by subtype of management in each treatment arm over the planned 52-week period. * Time to first HCP-managed VOC and duration of HCP-managed VOCs.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection evaluating treatment-emergent adverse events (TEAEs) along with real-time app tracking. **Serious Adverse Events (SAEs):** Monitoring for AEs of special interest including infusion-related reactions, serious infections, and opportunistic infections. **Data Monitoring Committee (DMC):** Independent External DMC to review blinded/unblinded safety data periodically.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population. **Sample Size Justification:** Target sample of $N = 315$ powered to detect a reduction in annualized VOC rate. **Handling of Missing Data:** Annualized VOC rate estimated with 95% confidence intervals; stratified Wilcoxon rank-sum test conducted using stratification factors (concomitant HU/HC usage and region).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring for centralized clinical and laboratory data. **Audit Trail:** eCRF locking, and rigorous data capture audit logs specifically tied to the novel cloud-based application used by patients to self-report VOCs.

11. Study Identifiers & Governance

Primary ID: CSEG101A2303 **Registry Number:** NCT06439082 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** SPARKLE **Official Regulatory Title:** A Phase III, Multicenter, Randomized, Placebo Controlled, Double-blind Study to Assess Efficacy and Safety of Crizanlizumab (5 mg/kg) Versus Placebo, With or Without Hydroxyurea/Hydroxycarbamide Therapy, in Adolescent and Adult Sickle Cell Disease Patients With Frequent Vaso-Occlusive Crises.

12. Clinical Framework (Sickle Cell Disease Specific)

Primary Condition: Sickle Cell Disease (SCD) **Diagnosis Threshold:** Confirmed SCD by Hb electrophoresis or HPLC, with 4-12 HCP-managed VOCs in prior 12 months. **Medical Methodology:** P-selectin blockade alongside standard of care (HU/HC). **Optimization Target:** Reduction of VOC frequency and mitigation of acute pain crisis events.

13. Study Procedures & Methodology

Management Pathway: Intravenous infusion schedule combined with standard outpatient SCD management. **Education Protocol (HCP):** Site initiation visits and continuous training on trial specific VOC definitions (≥ 4 hours treated as per local guidelines). **Education Protocol (Patient):** Comprehensive onboarding on utilizing the cloud-based application for real-time tracking of self-managed VOCs. **Quality Audit Mechanism:** Continuous digital

tracking & cross-referencing cloud-based app entries with standard eCRFs.

14. Observation & Follow-Up Schedule

Total Duration: 52-week treatment period **Onsite Visit Cadence:** Regular IV Infusion Schedule (typically Day 1, Week 2, and then Q4W/monthly) **Remote Monitoring Frequency:** Continuous via cloud-based application **Checklist Review Interval:** Safety and compliance review at each infusion visit

15. Sign-off & Approval

Role	Name	Signature	Date		:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					